

Review: Exponents and radicals



- 1 Explain how to evaluate 4^5 .
- 2 Explain how to determine if the answer is positive or negative when evaluating an expression with an integer power and negative base such as $(-7)^3$.
- 3 By considering the expressions $(-3)^2$ and -3^2 , state whether the each of the following statements is correct.
 - a Without parentheses, the base is -3 and the power is 2. The result is 9.
 - b If you have parentheses surrounding the -3 , then the base is 3 and the power is 2. You first evaluate to obtain 9, then take the negative, which is -9 .
 - c Without parentheses, the base is 3 and the power is 2. You first evaluate 3^2 to obtain 9, then take the negative, which is -9 .
 - d If you have parentheses surrounding the -3 , then the base is -3 and the power is 2. The result is 9.

- 4 $(-5)^{13}$ simplifies to which of the following?

A 5^{-13} **B** -5^{13} **C** -5^{-13} **D** 5^{13}

- 5 Evaluate:

- | | |
|---------------------------------|--|
| a 3^1 | b 1^3 |
| c 2^6 | d $(-2)^4$ |
| e $3^3 \times 3^2$ | f $(-4)^3$ |
| g $(-2)^3 \times 3^4$ | h -6^3 |
| i $7^2 - 2^2$ | j $(-11)^2 - (-9)^2$ |
| k $(-3)^2 \times (-2)^2$ | l -4^3 |
| m $(-3)^4$ | n -3^4 |
| o $(-4)^2$ | p $(-16)^2$ |
| q -15^2 | r $-(-8)^2$ |
| s $(-9) \times (-6)^2$ | t $(-2)^3 + 5^2$ |
| u $4 - (5 - 2)^2$ | v $6 - (12 - 3^2)$ |
| w $9 - 2^2 \times (-7)$ | x $(2 - 5)^2 \times (-21 + 14)$ |
| y $-(-3)^3 + 8^2$ | z $-(-4)^3 - (-10)^2 - 122$ |



6 A square patch of grass has a side length of 2^5 . Write an expression for the area of the grass in exponential form.

7 Determine whether each of the following statements is true or false.

- a $-x^n = (-x)^n$ for all x , only if n is odd.
- b $-x^n = (-x)^n$ for all x always.
- c $-x^n = (-x)^n$ for all x , only if n is even.

8 Evaluate the following:

- | | | | |
|----------------------------|--------------------------|----------------------|-----------------------|
| a $\sqrt{4}$ | b $\sqrt{16}$ | c $\sqrt{64}$ | d $\sqrt{100}$ |
| e $\sqrt{144}$ | f $\sqrt{25} - \sqrt{9}$ | g $\sqrt{8^2 + 6^2}$ | h $\sqrt{10^2 - 6^2}$ |
| i $\sqrt{144} - \sqrt{16}$ | j $\sqrt{16}$ | k $\sqrt{100}$ | l $\sqrt{169}$ |
| m $\sqrt{225}$ | n $-\sqrt{4}$ | o $-\sqrt{81}$ | p $-\sqrt{121}$ |
| q $-\sqrt{196}$ | | | |

Additional questions

9 For each of the following, determine the square root.

- a We know that $5 \times 5 = 25$. Find $\sqrt{25}$.
- b We know that $7 \times 7 = 49$. Find $\sqrt{49}$.
- c We know that $6 \times 6 = 36$. Find $-\sqrt{36}$.
- d We know that $7 \times 7 = 49$. Find $-\sqrt{49}$.

10 For each of the following, determine the cube root.

- a We know that $2 \times 2 \times 2 = 8$. What does $\sqrt[3]{8}$ equal?
- b We know that $4 \times 4 \times 4 = 64$. What does $\sqrt[3]{64}$ equal?
- c We know that $-3 \times -3 \times -3 = -27$. What does $\sqrt[3]{-27}$ equal?
- d We know that $-5 \times -5 \times -5 = -125$. What does $\sqrt[3]{-125}$ equal?

11 Evaluate the following:

- | | | | |
|------------------|-------------------|-------------------|--------------------|
| a $\sqrt[3]{8}$ | b $\sqrt[3]{64}$ | c $\sqrt[3]{125}$ | d $\sqrt[3]{27}$ |
| e $\sqrt[3]{-1}$ | f $\sqrt[3]{-27}$ | g $\sqrt[3]{-64}$ | h $\sqrt[3]{-125}$ |



12 Which is bigger: 3^3 or 5^2 ?

13 Evaluate the following expressions:

a $(2^2 + 3^3)$

b $(3 + 4)^2$

c $2^3 (3 + 7)$

d $(\sqrt{9} + 1)^3$

e $(\sqrt{3^2 + 4^2})^3$

f $\frac{2^3 + \sqrt{16}}{4}$

g $\sqrt{64} \div \sqrt[3]{8}$

h $2(\sqrt[3]{8} + 4)^2$

i $\left(\frac{1}{2}\right)^2 + \frac{1}{2}$

j $2\left(\frac{2}{3}\right)^2$

k $\left(\frac{2}{4} + \frac{1}{2}\right)^2$

l $\left(\frac{2}{3} + \sqrt{\frac{1}{9}}\right)^3$

m $\sqrt{1 - \left(\frac{6}{10}\right)^2}$

n $\left(-\frac{2}{3}\right)^2 + \sqrt{\frac{25}{81}}$

o $\sqrt{81} \div \sqrt[3]{125}$

p $2\left(\frac{1}{\sqrt[3]{8}} + 4\right)$

14 For each of the following situations:

- i** Determine the error each student made.
- ii** Write the correct work and answer.

- a** Ralph tried to evaluate the expression $2(3 + 2)^2$. He thought that he had the right answer and submitted the following to his teacher $2(3 + 2)^2 = 2(5)^2 = 10^2 = 100$.
- b** Ida tried to evaluate the expression $\sqrt{9}(8 \div 4 - 3 \times 4 + 4^2 + 3)$. She submitted the following to her teacher $\sqrt{9}(8 \div 4 - 3 \times 4 + 4^2 + 3) = \sqrt{9}(2 - 12 + 16 + 3) = \sqrt{9}(9) = \sqrt{81} = 9$.